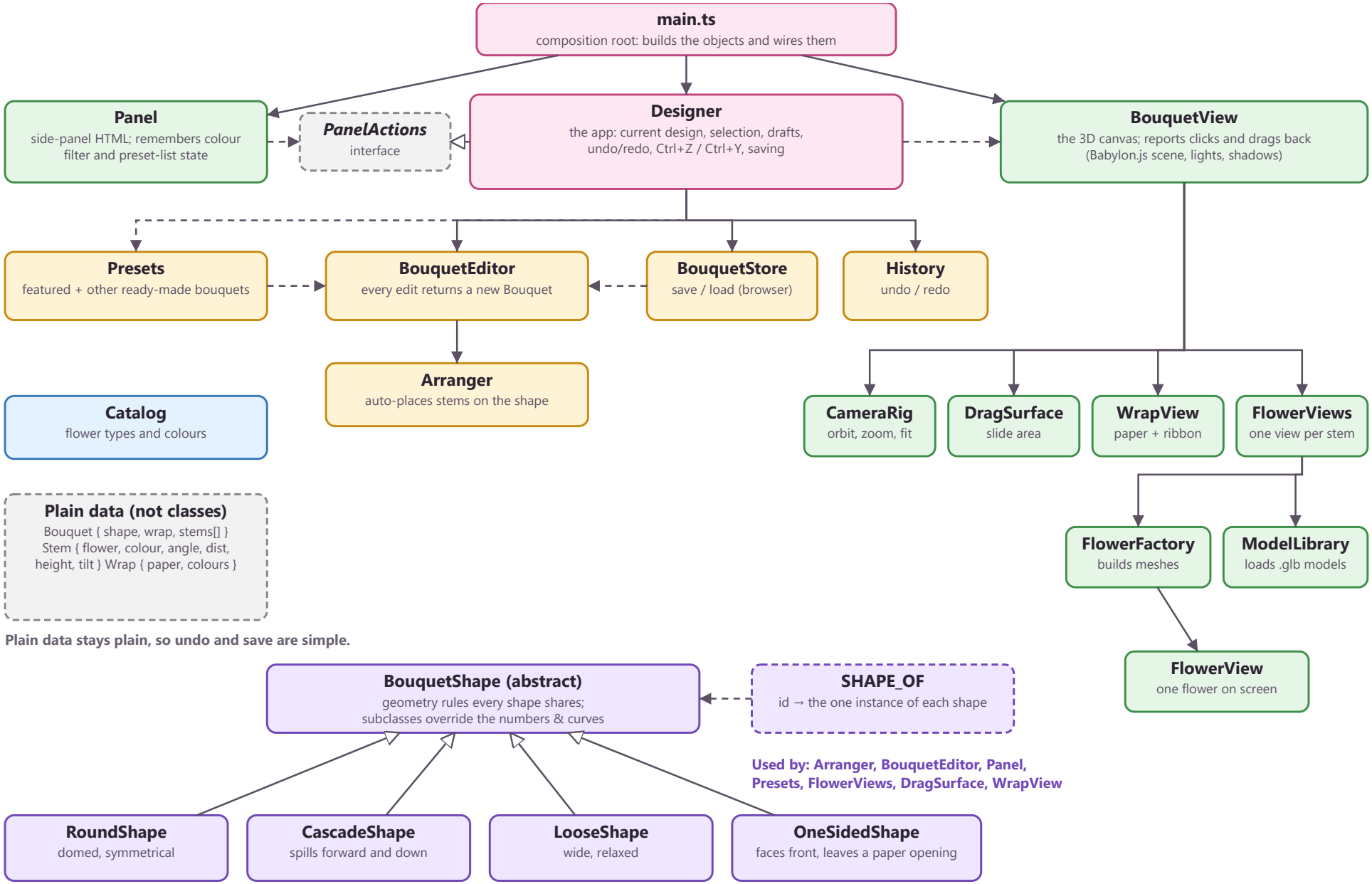
Bouquet Designer — class structure (OOP version)

Solid arrow = owns / creates · dashed arrow = uses · hollow triangle = extends / implements



What changed, in short

Same app, same look, same saved designs — the code underneath is now organised as classes.

Before → after

- **Before:** loose functions and file-level variables. `main.ts` held the app state and every action; shape differences were `if (shape === …)` checks in many places.
- **After:** each job has one class. `main.ts` is a 32-line composition root that builds the objects and connects them. The `Designer` holds the design, selection, undo/redo and keyboard shortcuts.
- **Inheritance:** the only real inheritance is `BouquetShape` and its four shapes. Each shape overrides just the numbers and curves that differ, so adding a fifth shape means adding one class.
- **Plain data stays plain:** `Bouquet`, `Stem` and `Wrap` are simple objects, so saving, undo and comparison work exactly as before.
- **Clean-up:** every class that holds resources has a `dispose()`. Tested, but the app does not call them yet.

How it runs

- A click in the 3D view → `BouquetView` calls back into `Designer` → the editor makes a new `Bouquet` → `History` records it → view and panel redraw and the store saves.
- Slider and drag previews only redraw the 3D view; one undo step is recorded when you let go.

Classes by part

Part	Classes
1 · data & editing	BouquetShape + 4 shapes, Catalog, Arranger, BouquetEditor, History, BouquetStore, Presets
2 · 3D view	ModelLibrary, FlowerFactory, FlowerView, FlowerViews, DragSurface, CameraRig, BouquetView, WrapView
3 · panel & app	Panel, Designer (implements PanelActions), main.ts wiring

How it was checked against the original

- TypeScript clean, production build OK, **170 / 170** tests pass (143 ported from the original + 27 new).
- Every placement number (15,632 lines) matches the original to the last bit.
- Both apps run side by side through the same browser steps: all 15 screenshots are byte-identical, as are screen positions and panel HTML (the camera-drag shot differs between any two runs, so it is excluded).
- Checked by hand in Chrome: front, side and back views, presets, select, drag, undo, saved design after reload, clean console.

Where things are

- Folder: `bouquet-designer-oop` (local git, not pushed). See the full change with `git diff c26b36f`.